

- [130] Aohan Zeng, Xiao Liu, Zhengxiao Du, Zihan Wang, Hanyu Lai, Ming Ding, Zhuoyi Yang, Yifan Xu, Wendi Zheng, Xiao Xia, et al. 2022. GLM-130B: An Open Bilingual Pre-trained Model. *arXiv preprint arXiv:2210.02414*.
- [131] Wei Zeng, Xiaozhe Ren, Teng Su, Hui Wang, Yi Liao, Zhiwei Wang, Xin Jiang, ZhenZhang Yang, Kaisheng Wang, Xiaoda Zhang, et al. 2021. PanGu-alpha: Large-scale Autoregressive Pretrained Chinese Language Models with Auto-parallel Computation. *arXiv preprint arXiv:2104.12369*.
- [132] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. 2022. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*.
- [133] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. 2023. A survey of large language models. *arXiv preprint arXiv:2303.18223*.
- [134] Chunting Zhou, Pengfei Liu, Puxin Xu, Srinu Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, et al. 2023. Lima: Less is more for alignment. *arXiv preprint arXiv:2305.11206*.
- [135] Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam Shazeer, and William Fedus. 2022. Designing effective sparse expert models. *arXiv preprint arXiv:2202.08906*.

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A Derivation of Data-Constrained Scaling Laws

Let N be the number of model parameters, D be the training tokens and U be the "unique" training tokens i.e. the size of the dataset that is to be trained on for one or more epochs. Chinchilla [42] only deals with non-repeated tokens, thus $D = U$ and we can write their formula ("Approach 3") as:

$$L(N, U) = \frac{A}{N^\alpha} + \frac{B}{U^\beta} + E \quad (7)$$

where E represents the irreducible loss. A , B , α and β are learned parameters.

We now want to generalize this expression to multiple epochs where tokens are repeated. We repeat the data R_D times, where $R_D = 0$ corresponds to the base case of a single epoch. We let D' be the "effective data size": the number of unique data needed to get the same value as repeating U unique tokens for R_D repeats. Hence, if $R_D = 0$, the effective data is the same as the total data processed. Intuitively, each time a sample is repeated, it is worth less as the model has already learned some of its information. Assume that each time a model trains on a token, it learns a $1 - \delta$ fraction of the information in it for some constant $0 \leq \delta \leq 1$. (Thus, if $\delta = 0$ repeated tokens are as good as new ones, and if $\delta = 1$, repeated tokens are worth nothing.) In other words, we expect the decrease in value of each repetition to be proportional to the value of the prior repetition, which is equivalent to exponential decay. As we would like to sum up the value of all repetitions, we temporarily assume an integral number of repeats and express it as a geometric series:

$$D' = U + (1 - \delta)U + (1 - \delta)^2U + \dots + (1 - \delta)^{R_D}U \quad (8)$$

We know that the sum S of a geometric series with a common ratio r is:

$$S = \frac{a(1 - r^n)}{1 - r} \quad (9)$$

where a is the first term and n the number of terms in the series. As $r = (1 - \delta)$ and $a = (1 - \delta)U$:

$$D' = U + U \sum_{k=1}^{R_D} (1 - \delta)^k = U + (1 - \delta)U \frac{(1 - (1 - \delta)^{R_D})}{\delta} \quad (10)$$

Note that Equation 10 can also be used with a non-integer number of repetitions. We can directly use Equation 10 as our effective data and learn δ but for convenience and interpretability, we redefine it in terms of the number of epochs beyond which repeating does not help. Note that as more data is repeated, the right-hand side tends to $\frac{(1 - \delta)U}{\delta}$, as $\lim_{R_D \rightarrow \infty} (1 - (1 - \delta)^{R_D}) = 1$. Let $R_D^* = \frac{1 - \delta}{\delta}$, hence D' "plateaus" at $U + R_D^*U$ as R_D goes to infinity.

If we assume δ to be small, $1 - \delta$ tends to one and we can approximate $1/R_D^* = \frac{\delta}{1 - \delta} \approx \delta$.

Next, define e^x in terms of its Taylor series expansion:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \approx 1 + x \quad (11)$$

If x is small later terms become increasingly small, thus $e^x \approx 1 + x$. As we have assumed δ to be small, let $x = -\delta$, which yields

$$(1 + x) = (1 - \delta) \approx e^{-\delta} \approx e^{-1/R_D^*} \quad (12)$$

Now inserting $(1 - \delta)/\delta = R_D^*$ and $(1 - \delta)^{R_D} = e^{(-1/R_D^*)R_D}$ into Equation 10 we get our final equation representing the *effective data*:

$$D' = U + U \cdot R_D^* \cdot (1 - e^{-R_D/R_D^*}) \quad (13)$$